

PHYS 143 – Mechanics and Thermal Physics

Newton's Laws and Forces

How do forces determine how an object moves?

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By the end of this week, you should be able to:

1. Identify forces and represent them using free-body diagrams.
2. Apply Newton's laws to equilibrium and accelerating systems.
3. Model normal force, tension and friction in physical systems.
4. Solve connected-object, inclined-plane and circular-dynamics problems.

Central question

How do forces determine how an object moves?

The thread of the week

interaction $\rightarrow$ **force** $\rightarrow$ **net force** $\rightarrow$ $\sum \vec{F} = m\vec{a} \rightarrow$ **motion**. Draw the free-body diagram first; then write $\sum \vec{F} = m\vec{a}$ by components.

Three contacts: 1 **REPRESENT** the forces, 2 **ANALYSE** how they determine acceleration, 3 **APPLY** them to contact forces and curved motion.

Reading: Young and Freedman, Chapter 4, §4.1–4.6 and Chapter 5, §5.1–5.4. **Labels:** **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use. More: [hhdonglo.github.io](https://github.com/hhdonglo).

Not marked. Answer individually, then check the answers.

1. A 50 N force acts 30° above the $+x$ axis. Find its components.
2. A car goes from rest to 20 m/s in 8.0 s. Find its acceleration.
3. Forces of 3.0 N east and 4.0 N north act on a body. Find the magnitude and direction of the resultant.
4. A ball is thrown upward. What are its velocity and acceleration at the top?
5. Solve $T - 20 = 2a$ and $30 - T = 3a$ for T and a .
6. An object moves in a circle of radius 2.0 m at 4.0 m/s. Find its acceleration and its direction.

Calculus is not needed this week. The Week 1 and Week 2 toolkits (components, signs, circular acceleration) are reused throughout.

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$$F_x = 50 \cos 30^\circ = 43.3 \text{ N}, \quad F_y = 50 \sin 30^\circ = 25 \text{ N}.$$

$$a = 20/8.0 = 2.5 \text{ m/s}^2.$$

$$R = \sqrt{3^2 + 4^2} = 5.0 \text{ N at } \tan^{-1}(4/3) = 53^\circ \text{ north of east.}$$

$$v = 0 \text{ (momentarily)}, \quad a = 9.8 \text{ m/s}^2 \text{ downward } (a_y = -g).$$

$$\text{Add: } 10 = 5a, \text{ so } a = 2.0 \text{ m/s}^2 \text{ and } T = 24 \text{ N.}$$

$$a = v^2/R = 16/2.0 = 8.0 \text{ m/s}^2, \text{ toward the centre.}$$

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Contact 1: Represent

What forces act, and how should we represent them?

Weeks 1 and 2: vectors $\rightarrow \vec{r} \rightarrow \vec{v} \rightarrow \vec{a}$.

How does an object move?

Week 3: interaction $\rightarrow$ force $\rightarrow$ net force $\rightarrow \vec{a}$.

Why does its motion change?

Opening question

A ball is thrown straight up. At its highest point, **which forces act on it, and what is the net force?**

Predict first

Poll: (a) none, (b) weight only, (c) weight and an upward “force of the throw”, (d) not enough information. Decide before the next click.

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top of flight: $v = 0$



(b) Weight only. Only one object interacts with the ball: the Earth, so the force is the **Earth on the ball**, its weight. There is no “force of the throw”: the hand stopped acting when the ball left it. The net force is mg downward, so $a = g$ downward (Week 2), even though $v = 0$ for an instant.

Quantity	What it is	What it measures	SI unit	Representation
Force	an interaction between two objects	how strongly, and in which direction, one object pushes or pulls another	newton, $1\text{ N} = 1\text{ kg m/s}^2$	an arrow on the free-body diagram, named by its agent
Net force	the vector sum of all forces on one object	the overall push that determines the acceleration	N	one arrow equal to the vector sum
Mass	the amount of matter in an object	how strongly it resists a change of velocity (inertia)	kg	a number; never drawn as an arrow
Weight	the gravitational force of the Earth on an object	how strongly the Earth pulls it	N	a downward arrow, $w = mg$

Interpret: a force answers *how strong, and in which direction?*; the net force tells us what the motion does; mass tells us how much it resists. Normal force, tension and friction are defined as they appear.

But what if... a force has no named source? Every force needs an **agent**: we **name** forces first.

Rule: naming a force

Every force is an interaction between **two** objects. Name it by who exerts it on whom:

$$\vec{F}_{A \text{ on } B} = \text{the force exerted by } A \text{ on } B$$

- A force is a **vector**: magnitude and direction. Unit: newton, $1 \text{ N} = 1 \text{ kg m/s}^2$.
- **Contact** forces: pushes, pulls, surfaces, ropes.
- **Long-range** forces: gravity (and electric and magnetic forces later).

Physical meaning

If you cannot say *which other object* exerts a force, it is not a force. This single habit will keep your free-body diagrams correct.

Force	Symbol	Agent on object	Direction
weight	$\vec{w}$	Earth on object	toward the Earth, $w = mg$
normal	$\vec{n}$	surface on object	perpendicular to the surface, away from it
tension	$\vec{T}$	rope on object	along the rope, pulling the object
friction	$\vec{f}$	surface on object	parallel to the surface, against the relative sliding
applied	$\vec{F}$	person or machine on object	as applied

At first, write the agent next to every force (“Earth on box”); later you can compress to w , n , T and f .

Rule: net force

Several forces on one object have the same effect as their **vector sum**:

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \cdots \quad \sum F_x = F_{1x} + F_{2x} + \cdots, \quad \sum F_y = F_{1y} + F_{2y} + \cdots$$

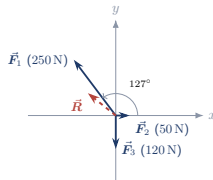
This is Week 1 vector addition by components: **resolve, combine, reconstruct**. Forces at an angle partly cancel, so add by components, not by magnitudes.

Example 1

Three forces act on a body: $F_1 = 250 \text{ N}$ at 127° from $+x$, $F_2 = 50 \text{ N}$ along $+x$, and $F_3 = 120 \text{ N}$ along $-y$. Find the net force.

Predict first

From the sketch: which quadrant is the net force in?



Identify: target $\sum \vec{F}$. **Set up:** x to the right, y up; angles from $+x$.

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	magnitude	x -component	y -component
$\vec{F}_1$	250 N	$250 \cos 127^\circ = -150 \text{ N}$	$250 \sin 127^\circ = +200 \text{ N}$
$\vec{F}_2$	50 N	+50 N	0
$\vec{F}_3$	120 N	0	-120 N
$\sum \vec{F}$		-100 N	+80 N

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Execute: $R = \sqrt{100^2 + 80^2} = 128 \text{ N}$; reference angle $\tan^{-1}(80/100) = 38.7^\circ$ in quadrant II, so $\theta = 141^\circ$ from $+x$.

Evaluate: quadrant II, as predicted; F_1 dominates. Units N. ✓

A sliding puck

A hockey puck is pushed along a rough floor and stops. On smooth ice it slides much farther. What would happen on a perfectly smooth surface?

The puck slows down because the **floor exerts a friction force on it**, not because motion *needs* a force. With less friction, the velocity changes less. With none, it does not change.

Rule: Newton's first law

If the net force on an object is zero, its velocity is constant: it stays at rest, or moves in a straight line at constant speed.

$$\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$$

Physical meaning

Inertia: the tendency to keep the same velocity. The law holds in an **inertial frame**, such as the ground or a train at constant velocity.

In which cases is $\sum \vec{F} = 0$?

1. A ball at the top of its flight.
2. A puck gliding at constant speed on frictionless ice.
3. A passenger sitting in a train moving smoothly at constant velocity.
4. A car going round a bend at constant speed.

Predict first

Hint: the first law concerns the *velocity vector*, not only the speed.

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Answer: (2) and (3). (1) $a = g$, so $\sum \vec{F} \neq 0$. (4) the direction of $\vec{v}$ changes, so $\vec{a} \neq 0$ and $\sum \vec{F} \neq 0$.

Common error: “no motion needs no force, and motion needs a force.” Motion needs no force; *changing* the velocity does.

Decide first

A passenger stands in a bus moving at 15 m/s . The bus brakes hard and the passenger lurches forward. Which force pushes the passenger forward?

Predict first

Draw the free-body diagram of the passenger: list the horizontal forces that actually act.

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None. The only horizontal force is the friction of the floor on the feet, directed *backward*: for a passenger of 70 kg and a deceleration of 4.0 m/s^2 it is $ma = 280 \text{ N}$. The upper body tends to keep its earlier velocity (first law), so relative to the bus it moves forward.

But what if the diagram needs a forward “inertial force” to explain the lurch? It does not: the bus is not an inertial frame, and in the ground frame the lurch needs no forward force.

- **Mass** m (kg): how much matter, and how strongly the object resists acceleration. The same everywhere.
- **Weight** w (N): the gravitational force of the Earth on the object. Weight depends on the place (g); mass does not.

Rule: weight

$$w = mg$$

directed toward the Earth's centre;
 $g = 9.8 \text{ m/s}^2$ near the Earth's surface;
units $\text{kg} \times \text{m/s}^2 = \text{N}$.

Example

A 2.0 kg object: weight on Earth ($g = 9.8 \text{ m/s}^2$) and on the Moon ($g = 1.62 \text{ m/s}^2$)?

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$w_{\text{Earth}} = 19.6 \text{ N}$, $w_{\text{Moon}} = 3.2 \text{ N}$; the mass is 2.0 kg in both places.

Free fall: $a = w/m = g$ for every mass, so a heavier object does not fall faster (Week 2).

Method: free-body diagram

1. **ISOLATE** one body and draw it as a dot.
2. **ASK** “Which other objects interact with this body?” Everything touching it, and the Earth.
3. **DRAW** one arrow per interaction, starting at the dot, in the direction the force acts *on the body*. Label each with its agent: “Earth on box (*w*)”, “table on box (*n*)”, and so on.
4. **AXES** are added afterwards (next contact).

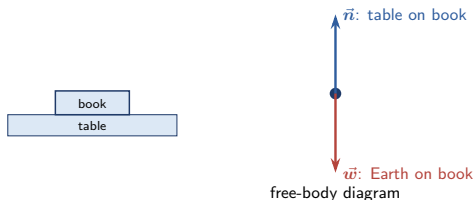
Rule

only forces acting *on* the body, each with a named agent.

Watch out

do not draw the net force, $m\vec{a}$, the velocity or forces the body exerts on other objects.

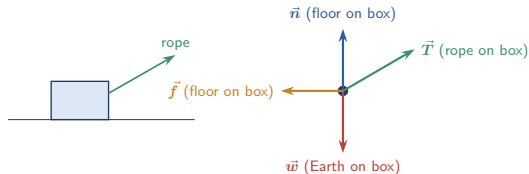
Check: can I name an agent for every arrow, and have I missed any object touching the body?



Predict first

If the book is at rest, how do n and w compare? Why?

$\sum \vec{F} = 0$ (first law), so $n = w$: the table pushes up exactly as hard as the Earth pulls down.



Four interactions, four forces

Earth on box ($\vec{w}$), floor on box ($\vec{n}$ and $\vec{f}$), rope on box ($\vec{T}$). The rope's force is angled, so it will have components.

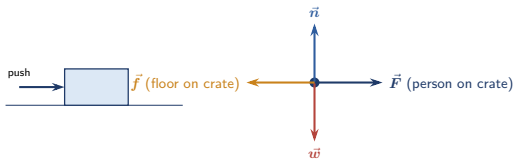
Independent problem

Draw a free-body diagram, with agent names, for each situation:

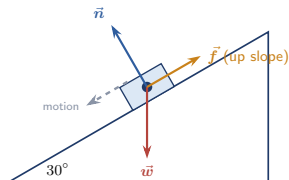
- (a) a crate pushed along a rough floor at **constant velocity**;
- (b) a block sliding down a **rough** incline.

Predict first

For (a): how do the push and the friction compare? For (b): in which direction does friction point?

**(a) A crate pushed at constant velocity**

Four forces: $\vec{w}$ (Earth on crate), $\vec{n}$ (floor on crate), $\vec{F}$ (person on crate), $\vec{f}$ (floor on crate). At constant velocity $\sum \vec{F} = 0$, so the push equals the friction, and $n = w$.

**(b) A block sliding down a rough incline**

Three forces: $\vec{w}$, $\vec{n}$ (perpendicular to the surface) and $\vec{f}$ (up the slope, against the sliding).

Common error

There is no separate “force down the slope”: that is a component of the weight, which we use in Contact 3.

Common mistakes and the correct habit

Mistake

Drawing a force without an agent, or drawing $\sum \vec{F}$, $m\vec{a}$ or $\vec{v}$ on the diagram.

Treating motion as needing a force.

Confusing mass with weight.

Adding force magnitudes.

Including forces the body exerts on other objects.

Instead

Only forces on the body, each named: “Earth on box”.

A net force changes the velocity; constant velocity needs none.

$w = mg$ depends on the place; mass does not.

Resolve, combine, reconstruct (Lecture 1).

Draw only what acts *on* the isolated body.

But what if... we can draw the forces, but not yet predict the motion? We need a law linking the **net force** to the **acceleration**.

Contact 2: Analyse

How do forces determine acceleration?

From representing forces to predicting motion

We can draw the forces on a body. **How do they determine its acceleration?** (Second law, equilibrium, third law, connected objects.)

Poll

The same 20 N horizontal push is applied to a 40 kg box and to an 80 kg box on a smooth floor. How do the accelerations compare?

Predict first

Equal? Half? Double? And how would the acceleration change if the push were doubled? Sketch the free-body diagram.

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Predict first

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The 80 kg box accelerates half as much. Doubling the force doubles the acceleration: $a \propto F$ and $a \propto 1/m$. For the 40 kg box: $\vec{w}$ and $\vec{n}$ are vertical and cancel, so $\sum F_x = 20 = 40 a_x$ and $a_x = 0.50 \text{ m/s}^2$, along the push (and 0.25 m/s^2 for 80 kg). ✓

But what if... the push is not horizontal, or several forces act? We need a law for the **net force**.

Rule: Newton's second law

$$\sum \vec{F} = m\vec{a} \quad \Longleftrightarrow \quad \sum F_x = ma_x, \quad \sum F_y = ma_y$$

The acceleration has the direction of the net force and is inversely proportional to the mass.

- $\sum \vec{F}$ means the net force: all forces *on this object*.
- Unit: $1 \text{ N} = 1 \text{ kg m/s}^2$. Forces cause *changes* of velocity, not velocity itself.
- Special case: $\vec{a} = 0$ gives equilibrium (first law).

Model assumptions

Valid in an inertial frame, for an object treated as a particle with constant mass. The momentum form $\sum \vec{F} = d\vec{p}/dt$ comes in a later lecture.

Decide first, and say which law you are using

1. Can an object move while the net force on it is zero?
2. Can the velocity be zero while the net force is not?
3. Is the acceleration always in the direction of the velocity?
4. If the net force on an object doubles, what happens to its acceleration? And if its mass doubles?

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Answers

1. **Yes.** Constant velocity: the first law.
2. **Yes.** A ball at the top of its flight: $\sum \vec{F} = m\vec{g}$.
3. **No.** The acceleration points along the net force; a car braking has $\vec{a}$ opposite to $\vec{v}$.
4. **It doubles; it halves.** $a = \sum F/m$.

Method: axes, components, equations

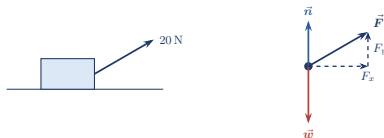
1. **FBD.** Draw the free-body diagram first (previous contact).
2. **CHOOSE** axes to simplify the physics, not automatically horizontal and vertical. Put one axis along the acceleration (or the motion) and the other perpendicular to it. On an incline: x parallel to the surface, y perpendicular.
3. **RESOLVE** each force that is not along an axis into components (Week 1): the angle from an axis, sketch the signs first. A component is positive along the positive axis direction, negative against it.
4. **WRITE** one equation per axis, $\sum F_x = ma_x$ and $\sum F_y = ma_y$, *before* inserting numbers. Count unknowns and equations; they must match. For equilibrium put $a = 0$.
5. **SOLVE** algebraically, then substitute with units. 6. **EVALUATE:** sign, units, size, a limiting case.

WATCH OUT: a force with no component along an axis does not appear in that axis equation; n is found from the equation perpendicular to the surface, so do not assume $n = mg$.

CHECK: do the components, recombined, give back the original force? Does the direction of $\vec{a}$ agree with the direction of the net force in the diagram?

Question

The same 40 kg box is now pulled by a rope with a 20 N force at 30° above the horizontal. The floor is still smooth. Find the acceleration and the normal force.

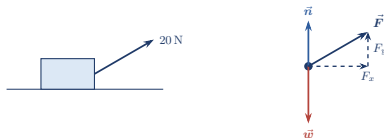


Predict first

Will a be larger or smaller than 0.50 m/s^2 ? Will n be larger or smaller than mg ?

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Predict first

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Set up: x horizontal along the motion, y up. Resolve $\vec{F}$: $F_x = 20 \cos 30^\circ = 17.3 \text{ N}$,
 $F_y = 20 \sin 30^\circ = 10.0 \text{ N}$.

Execute:

$$x: \sum F_x = F_x = ma_x \Rightarrow a_x = \frac{17.3}{40} = 0.43 \text{ m/s}^2$$

$$y: \sum F_y = n + F_y - mg = 0 \Rightarrow n = mg - F_y = 392 - 10.0 = 382 \text{ N} \quad (mg = 40 \times 9.8 = 392 \text{ N})$$

Evaluate: $a < 0.50$, because only part of the pull is horizontal; $n < mg$, because the rope lifts part of the weight. ✓

Common error

$n = mg$ only when no other force has a component perpendicular to the surface.

Rule: equilibrium

An object at rest, or moving at constant velocity, has $\sum \vec{F} = 0$:

$$\sum F_x = 0 \quad \sum F_y = 0$$

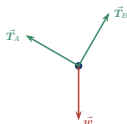
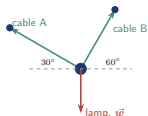
Equilibrium needs the *same* method: free-body diagram, axes, one equation per axis. The only difference is that the right-hand side is zero.

Predict first

A lamp hangs from two cables at different angles. Which cable carries more tension?

Question

A 10.0 kg lamp hangs at rest from two cables. Cable A makes 30° with the horizontal and cable B makes 60° on the other side. Find the tensions.



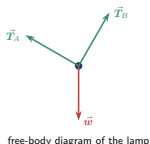
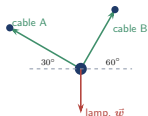
free-body diagram of the lamp

Predict first

Which cable, A or B, carries more tension? Check this against the answer.

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Predict first

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Set up: $w = 10.0 \times 9.8 = 98.0 \text{ N}$. x to the right, y up.
Components: T_A at 30° from $-x$, T_B at 60° from $+x$.

Execute:

$$x: -T_A \cos 30^\circ + T_B \cos 60^\circ = 0 \Rightarrow T_A = 0.577 T_B$$

$$y: T_A \sin 30^\circ + T_B \sin 60^\circ - 98.0 = 0$$

$$\text{Substitute: } (0.577)(0.500)T_B + 0.866 T_B = 98.0 \Rightarrow T_B = 84.9 \text{ N}, \quad T_A = 49.0 \text{ N}$$

Evaluate: cable B (steeper) carries more, as predicted.
Vertical check:

$$49.0 \sin 30^\circ + 84.9 \sin 60^\circ = 24.5 + 73.5 = 98.0 \text{ N}. \quad \checkmark$$

The tensions are not both $w/2$: unequal angles give unequal tensions.

Independent problem

A 1500 kg car speeds up from 10.0 m/s to 25.0 m/s in 6.0 s along a straight road. The resistive force on it is 400 N. Find the driving force exerted on the car by the road.

Predict first

Sketch the FBD. Is the driving force larger or smaller than the net force? Why?

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Week 2 kinematics: $a = (25.0 - 10.0)/6.0 = 2.5 \text{ m/s}^2$. Week 3 dynamics:

$\sum F_x = F - 400 = ma = 1500(2.5) = 3750 \text{ N}$, so $F = 4150 \text{ N}$.

Evaluate: larger than the net force, as predicted, because the resistive force must also be overcome. ✓

Rule: Newton's third law

If object A exerts a force on object B, then B exerts a force on A that is equal in magnitude and opposite in direction:

$$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$$

- The two forces are the **same type** and act at the same time.
- They act on **different bodies**, so they never appear in the same free-body diagram, and they never cancel.

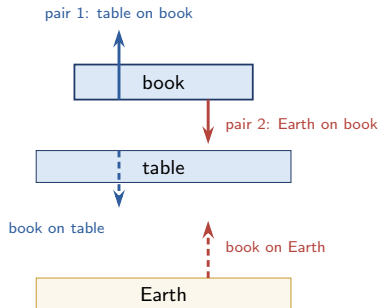
Finding the partner of a force

NAME it: "A on B". **FLIP** it: the partner is "B on A", the same type. **LOCATE** it: it acts on the *other* object.

COMPARE: equal and opposite; the accelerations differ if the masses differ. Two forces of equal size are not a pair unless they are "A on B" and "B on A".

Predict first

A small car and a heavy truck collide head-on. Which experiences the larger force? Which has the larger acceleration?

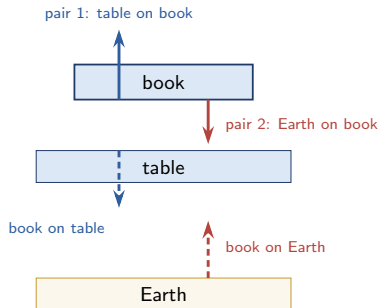


Question

For the book at rest, are $\vec{w}$ (Earth on book) and $\vec{n}$ (table on book) a third-law pair?

Predict first

Find the partner of $\vec{n}$. Of $\vec{w}$?



Question

For the book at rest, are $\vec{w}$ (Earth on book) and $\vec{n}$ (table on book) a third-law pair?

Predict first

Find the partner of $\vec{n}$. Of $\vec{w}$?

No. Both $\vec{w}$ and $\vec{n}$ act on the *book*. Partner of $\vec{n}$ (table on book): the force of the *book on the table*. Partner of $\vec{w}$ (Earth on book): the force of the *book on the Earth*.

Why are n and w equal here?

Because $\sum \vec{F} = 0$ on the book (first law), not because of the third law. If the table were in an accelerating lift, n and w would differ, but the pairs would still be equal and opposite.

Question

A horse pulls a cart forward. By the third law, the cart pulls back on the horse with an equal force. **Why can the horse and cart accelerate forward?**

Predict first

Draw the free-body diagram of the cart, and then of the horse, separately.

Question

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Predict first

Draw the free-body diagram of the cart, and then of the horse, separately.

Cart: the forward pull by the horse is greater than the resistive forces, so it accelerates. **Horse:** the forward force on its hooves from the *ground* is larger than the backward pull from the cart. The pair (horse on cart, cart on horse) acts on different bodies and does not cancel in either diagram.

A heavy truck and a small car in a collision exert equal forces on each other; the car accelerates more because $a = F/m$.

Rule: ideal rope and pulley

An **ideal rope** is light and does not stretch: it has the same tension throughout, and objects joined by a taut rope have the same acceleration magnitude. An **ideal pulley** is light and frictionless and only changes the direction of the rope.

Physical meaning

Real ropes have mass and stretch, and real pulleys have friction. The ideal model is valid when these effects are small.

Common error

The tension is not automatically equal to a weight. It depends on the acceleration.

Method: connected objects

1. **ISOLATE** each body and draw its own free-body diagram.
2. **CONSTRAINT**: state the link: same acceleration magnitude, same rope tension (ideal rope and pulley).
3. **AXES**: choose each body's axis along its own motion, with signs consistent with the constraint.
4. **WRITE** one $\sum F = ma$ equation per body along its motion.
5. **SOLVE** the simultaneous equations for a and T .

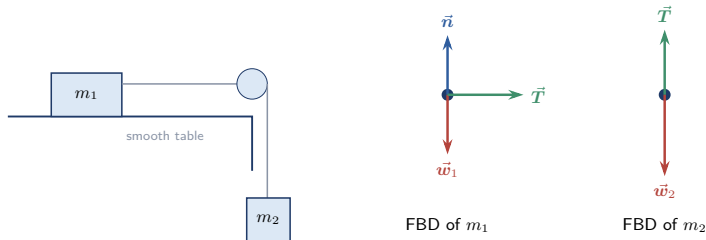
Watch out

drawing one diagram for the whole system and forgetting that internal tensions appear only when you separate the bodies.

Check

treat the whole system once: the external force equals the total mass times a . Check limiting cases.

A block $m_1 = 3.0 \text{ kg}$ on a smooth table is joined by an ideal rope over an ideal pulley to a hanging block $m_2 = 2.0 \text{ kg}$. Released from rest, find the acceleration and the tension.



Predict first

Is T larger or smaller than $m_2g = 19.6 \text{ N}$? Is a larger or smaller than g ?

Set up: m_1 : $+x$ toward the pulley; m_2 : $+y$ downward (consistent with the constraint). The same a and T for both.

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Execute:

$$m_1 (x): T = m_1 a \quad m_2 (y, \text{down}): m_2 g - T = m_2 a$$

$$\text{Add: } m_2 g = (m_1 + m_2) a \Rightarrow a = \frac{m_2 g}{m_1 + m_2} = \frac{19.6}{5.0} = 3.9 \text{ m/s}^2; \quad T = m_1 a = 11.8 \text{ N}$$

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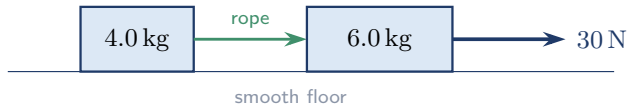
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Evaluate: $T < m_2 g$ and $a < g$, as predicted. Whole-system check: the only external horizontal driving force is $m_2 g$, so $m_2 g = (m_1 + m_2) a$. ✓

Limits: $m_1 \rightarrow 0$ gives $a = g$, $T = 0$ (free fall); $m_2 \rightarrow 0$ gives $a = 0$.

Independent problem

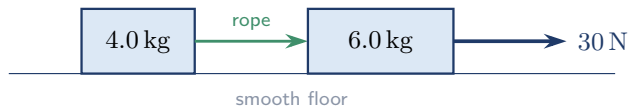
Two blocks on a smooth floor are joined by an ideal rope and pulled by a horizontal 30 N force as shown. Find the acceleration and the tension in the rope.

**Predict first**

Is T larger or smaller than 30 N? Draw both free-body diagrams.

Independent problem

Two blocks on a smooth floor are joined by an ideal rope and pulled by a horizontal 30 N force as shown. Find the acceleration and the tension in the rope.



Predict first

Is T larger or smaller than 30 N? Draw both free-body diagrams.

Whole system: $a = \frac{30}{4.0 + 6.0} = 3.0 \text{ m/s}^2$. The 4.0 kg block: $T = m_1 a = 12 \text{ N}$. Check on the 6.0 kg block: $30 - 12 = 18 = 6.0(3.0)$. ✓ $T < 30 \text{ N}$, as predicted.

Common mistakes and the correct habit

Mistake**Instead**

Assuming $n = mg$ on every surface.

Find n from the equation perpendicular to the surface.

Treating a third-law pair as two forces on one body.

The pair acts on *different* bodies, so it never cancels.

Assuming the tension equals a weight.

Write $\sum F = ma$ for each body; T depends on a .

Always using horizontal and vertical axes.

Put one axis along the acceleration or the slope.

Putting a force into an axis along which it has no component.

A perpendicular force does not appear in that equation.

But what if... every surface so far was smooth, and every path straight? Real surfaces exert **friction**, and objects also move on **curved paths**. Contact 3 extends the model.

Contact 3: Apply

How do contact forces and curved motion extend the model?

From ropes and smooth floors to real surfaces

Real surfaces exert friction, and real paths curve. **How do contact forces and curved motion extend the model?**

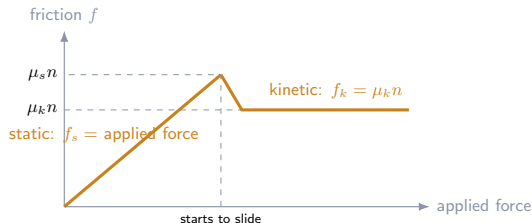
This contact: **friction; inclined planes; circular dynamics**; one integrated problem.

Poll

You push a heavy crate. Nothing happens. You push harder: still nothing. Harder again, and it suddenly slides. How does the floor's friction force change as you push harder?

Predict first

Sketch friction against applied force before the next slide.



Rule: friction

Static (no sliding): $0 \leq f_s \leq \mu_s n$

Kinetic (sliding): $f_k = \mu_k n$

Direction: parallel to the surface, **against the relative sliding** (or the tendency to slide) of the two surfaces. Usually $\mu_k < \mu_s$; μ has no units.

$f_s = \mu_s n$ is the maximum static friction, not its value

Static friction adjusts itself to match the other forces, up to $\mu_s n$. Two ways to get it wrong: (1) using $f_s = \mu_s n$ when the object is not about to slide; (2) using $\mu_s n$ when it is already sliding: then use $f_k = \mu_k n$.

Question

A 40.0 kg crate rests on a level floor with $\mu_s = 0.50$ and $\mu_k = 0.40$. (a) A horizontal push of 150 N is applied: does the crate move, and what is the friction? (b) The push is increased to 250 N: find the acceleration.

Predict first

Sketch the FBD. Estimate: which push is likely to exceed the maximum static friction?

Method: friction

1. NORMAL FORCE: find n from the equation perpendicular to the surface; do not assume $n = mg$. **2. IS IT SLIDING?** If not, friction is whatever is needed (static), and must satisfy $f_s \leq \mu_s n$. **3. IF SLIDING,** $f_k = \mu_k n$, against the relative sliding. **4.** Write $\sum F = ma$ along the motion. Friction opposes sliding relative to the *surface*, not always the motion relative to the ground.

Set up: level floor, no vertical acceleration, so $n = mg = 40.0 \times 9.8 = 392 \text{ N}$. Maximum static friction: $\mu_s n = 0.50 \times 392 = 196 \text{ N}$.

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(b) $250 > 196 \text{ N}$: it slides. $f_k = \mu_k n = 0.40 \times 392 = 157 \text{ N}$; $\sum F_x = 250 - 157 = 93 \text{ N}$;
 $a = 93/40.0 = 2.3 \text{ m/s}^2$.

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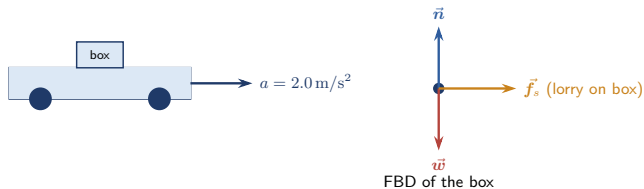
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Evaluate: (a) $f_s \leq \mu_s n$ ✓. (b) once the crate slides, the friction drops from 196 N to 157 N, so the net force is positive. ✓

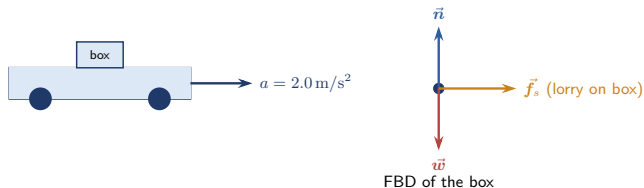
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The **static friction** of the lorry on the box points forward, along the box's motion: $f_s = ma$. It requires $\mu_s \geq a/g = 0.20$.

Friction opposes the *sliding* between surfaces; it can point along the motion relative to the ground.

Decide first

A car accelerates forward from rest on a level road. Which force on the car accelerates it? (a) the force of the engine, (b) the friction of the road on the tyres, (c) the force of the tyres on the road, (d) the weight.

Predict first

Hint: the engine and the wheels are parts of the car. Which *other* object pushes the car?

Decide first

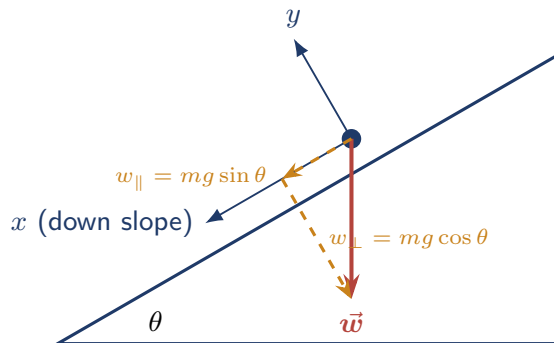
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Predict first

Hint: the engine and the wheels are parts of the car. Which *other* object pushes the car?

(b). The driven tyres push the road backward; by the third law the road pushes the tyres forward with **static friction**. Forces inside the car cannot change the motion of the car as a whole. (c) acts on the road, not on the car.

But what if the road is icy? Static friction is limited: $a_{\max} = \mu_s g$, so $\mu_s = 0.80$ allows at most 7.8 m/s^2 , and a small μ_s allows almost none however powerful the engine.

**Rule: weight on an incline**

Axes: x along the slope, y perpendicular.

$$w_{\parallel} = mg \sin \theta, \quad w_{\perp} = mg \cos \theta$$

EQUATION – components of weight

Meaning. the weight resolved along and across the slope.

Variables, units. θ slope angle; m , g as before; N.

Interpretation. $\theta \rightarrow 0$: $w_{\parallel} \rightarrow 0$;
 $\theta \rightarrow 90^\circ$: $w_{\perp} \rightarrow 0$.

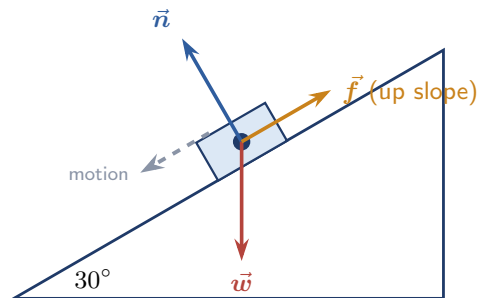
Example. 5.0 kg, 30° : 24.5 N and 42.4 N.

Question

A 5.0 kg block slides down a 30.0° incline with kinetic friction $\mu_k = 0.30$. Find its acceleration.

Predict first

Is a larger or smaller than $g \sin 30^\circ = 4.9 \text{ m/s}^2$? Does the answer depend on the mass?



Set up: x down the slope (the direction of motion), y perpendicular. Forces: $\vec{w}$ (components $mg \sin \theta$ along $+x$, $mg \cos \theta$ along $-y$), $\vec{n}$ along $+y$, $\vec{f}$ along $-x$.

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Execute:

$$y: n - mg \cos \theta = 0 \Rightarrow n = 5.0(9.8) \cos 30^\circ = 42.4 \text{ N}$$

$$f_k = \mu_k n = 0.30 \times 42.4 = 12.7 \text{ N}$$

$$x: mg \sin \theta - f_k = ma \Rightarrow 24.5 - 12.7 = 5.0 a \Rightarrow a = 2.4 \text{ m/s}^2$$

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Evaluate: $a < 4.9 \text{ m/s}^2$, as predicted. Algebra: $a = g(\sin \theta - \mu_k \cos \theta)$, so the mass cancels.

With $\mu_k = 0$ this gives $g \sin \theta$. ✓

Poll

A car goes round a flat circular bend at a **constant speed**. Is the net force on it zero? If not, in which direction does it point?

Predict first

Recall from Week 2: what is the direction and size of the acceleration in uniform circular motion?

But what if... the acceleration v^2/R toward the centre is known, but *which force* supplies it? We need the dynamics of circular motion.

Rule: net force in uniform circular motion

The acceleration is $a_{\text{rad}} = v^2/R$ toward the centre (Week 2), so the net force must point toward the centre:

$$\sum F_{\text{radial}} = m \frac{v^2}{R} = m\omega^2 R \quad (\text{kg m}^2/\text{s}^2/\text{m} = \text{N})$$

“Centripetal force” is not a new kind of force

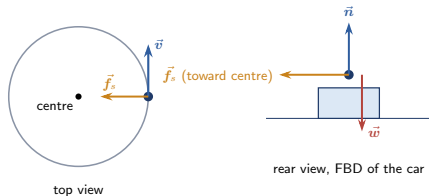
It is the name for the *net inward force*, supplied by real forces: friction, tension, the normal force, gravity. It never appears as an extra arrow on a free-body diagram.

Method: circular dynamics

1. FBD with real forces only.
2. Radial axis positive toward the centre.
3. Write $\sum F_{\text{radial}} = mv^2/R$ and the perpendicular equation (flat curve: $n = mg$).
4. Identify which real force supplies the inward net force.
5. The maximum speed occurs when that force reaches its maximum value.

Question

A 1200 kg car rounds a flat circular bend of radius 50.0 m at 15.0 m/s. (a) What friction force is required, and what is the least μ_s ? (b) What is the maximum speed for $\mu_s = 0.80$ (dry) and $\mu_s = 0.30$ (wet)?



Predict first

Which force points toward the centre? Does the answer depend on the mass?

Set up: radial axis toward the centre; the only horizontal force is static friction, so f_s supplies the net inward force. Vertical: $n = mg = 11\,760\text{ N}$.

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$$(a) \quad f_s = \frac{mv^2}{R} = \frac{1200(15.0)^2}{50.0} = 5400\text{ N}; \quad \mu_{s,\min} = \frac{f_s}{n} = 0.46.$$

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(b) At the limit $f_s = \mu_s mg = mv^2/R$, so $v_{\max} = \sqrt{\mu_s g R}$ (the mass cancels, so $v_{\max}$ does not depend on the mass; a wet road lowers it).

Dry: $\sqrt{0.80 \times 9.8 \times 50.0} = 19.8\text{ m/s}$. Wet: $\sqrt{0.30 \times 9.8 \times 50.0} = 12.1\text{ m/s}$.

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Evaluate: 15.0 m/s is safe when dry and unsafe when wet, which matches everyday experience. ✓

In the car, you feel thrown outward.

Is there an outward force pushing you?

In the ground (inertial) frame, **no outward force acts on you**. Your body tends to move in a straight line (first law), while the car turns toward the centre.

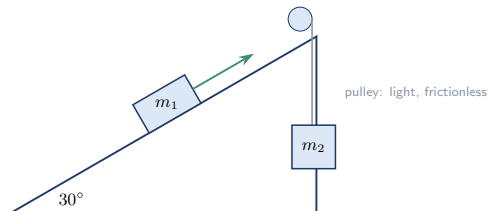
What actually happens

The seat and door exert a force on you toward the centre. That inward net force is what makes you follow the curve; the “outward push” is your inertia.

Application problem: decide what physics applies

A 10.0 kg crate rests on a rough 30° incline ($\mu_s = 0.30$, $\mu_k = 0.25$). A light rope runs from the crate, parallel to the slope, over a frictionless pulley to a 8.0 kg hanging block. The system is released from rest.

Does the system move? If so, in which direction? Find the acceleration and the tension.



Predict first

IDENTIFY $\rightarrow$ FBDs and axes $\rightarrow$
 PREDICT $\rightarrow$ EXECUTE $\rightarrow$
 EVALUATE.

Identify, set up: two bodies, connected objects, a slope and friction: which tool, and why?
Axes: along the slope for the crate (positive up the slope), vertical for the block (positive down), consistent with the constraint.

Identify, set up: two bodies, connected objects, a slope and friction: which tool, and why?
Axes: along the slope for the crate (positive up the slope), vertical for the block (positive down), consistent with the constraint.

Does it move? Net pull toward the block: $m_2g - m_1g \sin 30^\circ = 78.4 - 49.0 = 29.4 \text{ N}$.
Maximum static friction: $\mu_s n = 0.30 \times (10.0)(9.8) \cos 30^\circ = 25.5 \text{ N}$. Since $29.4 > 25.5$, it moves, with the crate going **up** the slope.

Identify, set up: two bodies, connected objects, a slope and friction: which tool, and why?
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Execute: $f_k = 0.25 \times 84.9 = 21.2 \text{ N}$ (down the slope). Crate: $T - 49.0 - 21.2 = 10.0 a$.
Block: $78.4 - T = 8.0 a$.

Add: $8.2 = 18.0 a \Rightarrow a = 0.455 \approx 0.46 \text{ m/s}^2$; $T = 78.4 - 8.0(0.455) = 74.8 \text{ N}$.

Identify, set up: two bodies, connected objects, a slope and friction: which tool, and why?
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Evaluate: $T < m_2g = 78.4 \text{ N}$; small a , because the pull barely exceeds the resistance;
whole-system check $m_2g - m_1g \sin 30^\circ - f_k = (m_1 + m_2)a$. ✓

Common mistakes and the correct habit

Mistake**Instead**

Using $f_s = \mu_s n$ when the object is not about to slide.

$f_s \leq \mu_s n$; it adjusts to the other forces.

Believing friction always opposes the motion relative to the ground.

It opposes sliding relative to the surface.

Adding a “centripetal” or “centrifugal” arrow to the diagram.

The inward net force is supplied by real forces.

Drawing a separate “force down the slope”.

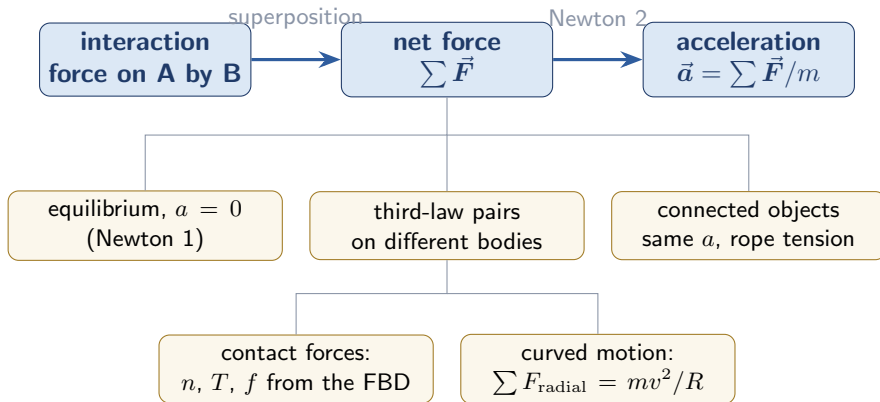
It is a component of the weight.

Swapping $\sin \theta$ and $\cos \theta$ on a slope.

Test the limits $\theta = 0$ and $\theta = 90^\circ$.

Letting the mass decide $v_{\max}$ on a curve.

$v_{\max} = \sqrt{\mu_s g R}$: the mass cancels.



Key ideas

- Every force is an interaction between two objects and has a named agent; a free-body diagram contains only the forces on the isolated body.
- Net force is the vector sum of the forces on one object; mass is the resistance to acceleration, weight $w = mg$ is a force and depends on the place.
- $\sum \vec{F} = m\vec{a}$ holds component by component; with $\sum \vec{F} = 0$ the velocity is constant (first law).
- Choose axes along the acceleration or the surface; n comes from the perpendicular equation, not from $n = mg$.
- Third-law pairs are equal and opposite and act on different bodies, so they never cancel.
- Connected objects: one free-body diagram and one equation per body, linked by the constraint (same a , same T for an ideal rope).
- $f_s \leq \mu_s n$ is a limit and adjusts to the other forces; $f_k = \mu_k n$ is fixed; friction opposes sliding relative to the surface.
- Uniform circular motion needs an inward net force mv^2/R supplied by real forces; there is no “centrifugal” or “centripetal” arrow.
- Habit: sketch, predict, calculate, evaluate.

Reference tools: the next slides (master toolkit, core equations). The Reference deck holds the seven toolkits, the equation-anatomy slides and the master procedure.

MASTER TOOLKIT – Force problems

1. **IDENTIFY** Which body or bodies, and what is asked?
2. **ISOLATE** One body at a time.
3. **DRAW FBD(S)** Forces on the body, each named by agent.
4. **CHOOSE AXES** Along the acceleration or the surface.
5. **PREDICT** Signs, direction, trend, rough size.
6. **SELECT MODEL** Equilibrium? Accelerating? Pair? Connected? Friction? Incline? Circular?
7. **EXECUTE** $\sum F = ma$ by components, one equation per axis per body; key move (right); algebra first, then numbers.
8. **EVALUATE** Units, sign, direction, limits, back to the FBD.
Result disagrees with the prediction? Investigate.
Seven toolkits: Reference deck.

Step 7: the key move for each model

FBD: only forces on the body, each with an agent.

Axes: one axis along the acceleration or slope; resolve the rest.

Second law: $\sum F_x = ma_x$, $\sum F_y = ma_y$; n from the perpendicular equation; equilibrium is $a = 0$.

Third law: name, flip, locate; pairs act on different bodies.

Connected: one equation per body; same a and T .

Friction: sliding? $f_s \leq \mu_s n$ or $f_k = \mu_k n$.

Incline: $w_{\parallel} = mg \sin \theta$, $w_{\perp} = mg \cos \theta$.

Circular: $\sum F_{\text{radial}} = mv^2/R$ toward the centre; real forces only.

Framework: IDENTIFY (1–2), SET UP (3–6), EXECUTE (7), EVALUATE (8).

Idea	Equation	Conditions and units
Net force	$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$	vector sum; N
First law	$\sum \vec{F} = 0 \Rightarrow \vec{a} = 0$	inertial frame
Second law	$\sum F_x = ma_x, \quad \sum F_y = ma_y$	particle of constant mass; N
Third law	$\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$	different bodies
Weight	$w = mg$	$g = 9.8 \text{ m/s}^2$ near Earth
Friction	$0 \leq f_s \leq \mu_s n, \quad f_k = \mu_k n$	μ dimensionless
Incline	$w_{\parallel} = mg \sin \theta, \quad w_{\perp} = mg \cos \theta$	θ from the horizontal
Circular	$\sum F_{\text{radial}} = mv^2/R = m\omega^2 R$	toward the centre
Flat curve	$v_{\text{max}} = \sqrt{\mu_s g R}$	friction supplies the force

Forces and the free-body diagram

The forces on your diagram decide which ones do work (Lecture 4), how momentum changes (Lecture 7), and which produce torque (Lecture 8).

Second law and energy

Energy methods (Lectures 4 and 5) give speeds without solving for a ; the momentum form $\sum \vec{F} = d\vec{p}/dt$ is the general second law (Lecture 7).

Friction

Friction removes mechanical energy and warms the surfaces: work and energy in Lectures 4 and 5, heat and thermal energy in Lectures 9 to 11.

Inclines and ramps

The same slope problem is solved much more simply by energy conservation in Lecture 5, with no free-body axes.

Circular dynamics

$\sum F_{\text{radial}} = mv^2/R$ becomes the dynamics of rotating bodies in Lecture 8: torque and angular acceleration.

Physical meaning

The free-body diagram and component equations are the method you will keep: later topics change *which* forces appear, not how you set up the problem.

- ☐ name a force by who exerts it on whom?
- ☐ find a net force by components?
- ☐ distinguish mass, weight and normal force?
- ☐ draw a correct free-body diagram?
- ☐ apply the first and second laws by components?
- ☐ solve equilibrium problems?
- ☐ identify third-law pairs and explain why they do not cancel?
- ☐ solve connected-object problems with ideal ropes and pulleys?
- ☐ apply $f_s \leq \mu_s n$ and $f_k = \mu_k n$ with the correct direction?
- ☐ solve incline problems with sensible axes?
- ☐ explain and apply $\sum F_{\text{radial}} = mv^2/R$?
- ☐ select the appropriate model in an unfamiliar problem?

Next

Tutorial sheet: the lift, extra FBD, connected-object, friction, incline and circular problems, and the banked curve, conical pendulum and vertical circle. The Reference deck holds the supporting material cut from the live lecture; the supplementary notes hold inertial frames, the momentum form of the second law, drag and terminal speed, and the four fundamental forces.